Lecture 11:
From CNN to HPC: Why Modern AI Needs High-Performance Computing?

Er.Roshan Shrestha
roshan.shrestha@islingtoncollege.edu.np

Agenda

- CNN
- HPC

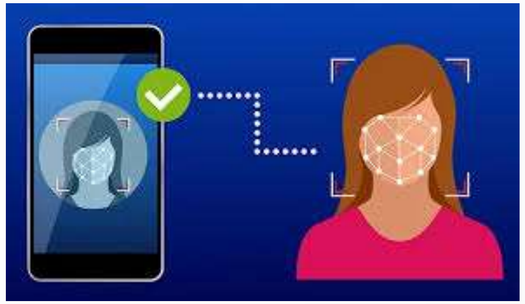
Milestone 1 (Wednesday, 7 January 2026)

- **Introduction**
  - *Explanation of the topic/AI concepts used*
  - *Explanation/introduction of the chosen problem domain/topic*
- **Background**
  - *Research work done in coursework 1*
- **Solution**
  - *Explanation of the solution/used AI algorithm*
  - *Pseudocode of the solution*
  - *Diagrammatical representations of the solution (flowcharts/state transition diagrams)*
  - *Explanation of the development process (with explanation of the used tools and technologies/libraries)*
  - *Achieved results (screenshots of the application/screenshots of the results attained)*
- ***Application***
  - *Developed application that runs (pre – compiled if required) including source code and any other required files*

Unlocking the phone

"How does your phone unlock your face faster than your friend recognizes you?"

Your phone doesn't see *you*.
It sees patterns of pixels, and CNNs are experts at spotting patterns.



1 Unlocking the phone
2 Recognizing the images
3 Issue of normal neural network
4 CNN
5 Standard Classification of images
6 Types of Images
7 Colored image
8 Understanding the pixel grid
9 Edge Detection
10 Edge Detection importance
11 Detecting edges
12 Understanding Filter
13 Vertical Edge Detection
14 Horizontal Edge Detection
15 Convolution Operation
16 Kernel
17 Stride and Padding
18 Pooling Layer
19 CNN Architecture
20 Vertical Edge Detection
21 Detection
22 Resultant image
23 Filters
24 What Else Can Convolution Do?
25 How CNNs Use Layers
26 Applications
27 Training CNN
28 Famous CNN Architectures
29 CNN vs ANN vs RNN
30 HPC
Distributed memory
32 Data Parallelism
33 Model Parallelism
34 Pipeline Parallelism
35 Hybrid Parallelism

Recognizing the images

How do YOU recognize them?

Shape
Edges
Eyes
Texture

CNN learns exactly these things — layer by layer

1 Unlocking the phone
2 Recognizing the images
3 Issue of normal neural network
4 CNN
5 Standard Classification of images
6 Types of Images
7 Colored image
8 Understanding the pixel grid
9 Edge Detection
10 Edge Detection importance
11 Detecting edges
12 Understanding Filter
13 Vertical Edge Detection
14 Horizontal Edge Detection
15 Convolution Operation
16 Kernel
17 Stride and Padding
18 Pooling Layer
19 CNN Architecture
20 Vertical Edge Detection
21 Detection
22 Resultant image
23 Filters
24 What Else Can Convolution Do?
25 How CNNs Use Layers
26 Applications
27 Training CNN
28 Famous CNN Architectures
29 CNN vs ANN vs RNN
30 HPC
Distributed memory
32 Data Parallelism
33 Model Parallelism
34 Pipeline Parallelism
35 Hybrid Parallelism

Issue of normal neural network in case of Images

2D Image (64×64)
↓ flatten
1D Vector (4096)

Fully connected NN:

A face on the left vs right looks the same after flattening

Images have structure
Nearby pixels are related.

CNN preserves spatial relationships.

Number of weights
=4096×1000=4,096,000 parameters
This is only one layer

1 Unlocking the phone
2 Recognizing the images
3 Issue of normal neural network
4 CNN
5 Standard Classification of images
6 Types of Images
7 Colored image
8 Understanding the pixel grid
9 Edge Detection
10 Edge Detection importance
11 Detecting edges
12 Understanding Filter
13 Vertical Edge Detection
14 Horizontal Edge Detection
15 Convolution Operation
16 Kernel
17 Stride and Padding
18 Pooling Layer
19 CNN Architecture
20 Vertical Edge Detection
21 Detection
22 Resultant image
23 Filters
24 What Else Can Convolution Do?
25 How CNNs Use Layers
26 Applications
27 Training CNN
28 Famous CNN Architectures
29 CNN vs ANN vs RNN
30 HPC
Distributed memory
32 Data Parallelism
33 Model Parallelism
34 Pipeline Parallelism
35 Hybrid Parallelism

CNN

- CNN is a deep learning model that automatically learns visual features from images using convolution operations.
- *CNN is a deep learning model designed to efficiently learn spatial patterns in image data.*

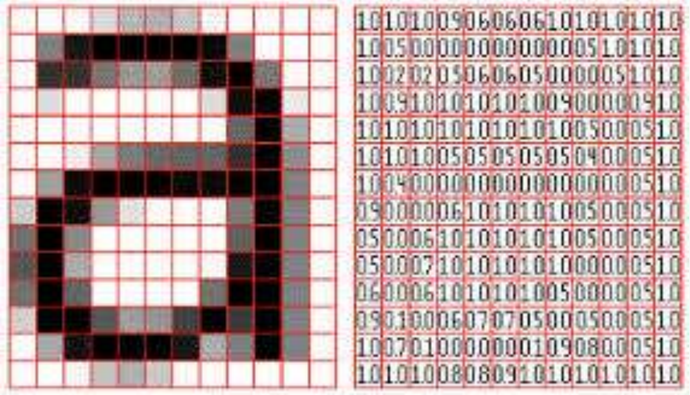
Human Vision vs CNN

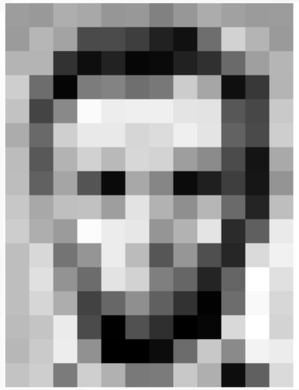
Human Brain	CNN
Eyes detect edges	Conv layer
Brain detects shapes	Deeper conv
Recognizes object	Dense layer

1 Unlocking the phone
2 Recognizing the images
3 Issue of normal neural network
4 CNN
5 Standard Classification of images
6 Types of Images
7 Colored image
8 Understanding the pixel grid
9 Edge Detection
10 Edge Detection importance
11 Detecting edges
12 Understanding Filter
13 Vertical Edge Detection
14 Horizontal Edge Detection
15 Convolution Operation
16 Kernel
17 Stride and Padding
18 Pooling Layer
19 CNN Architecture
20 Vertical Edge Detection
21 Detection
22 Resultant image
23 Filters
24 What Else Can Convolution Do?
25 How CNNs Use Layers
26 Applications
27 Training CNN
28 Famous CNN Architectures
29 CNN vs ANN vs RNN
30 HPC
Distributed memory
32 Data Parallelism
33 Model Parallelism
34 Pipeline Parallelism
35 Hybrid Parallelism

Recognizing the images

An image is represented as a **matrix of numbers (pixels)**





157	153	174	168	150	152	129	151	172	161	155	156
155	182	163	74	75	62	33	17	110	210	180	154
180	180	50	14	34	6	10	33	48	106	159	181
206	109	5	124	131	111	120	204	166	15	56	180
194	68	137	251	237	239	239	228	227	87	71	201
172	105	207	233	233	214	220	239	228	98	74	206
188	88	179	209	185	215	211	158	139	75	20	169
189	97	165	84	10	168	134	11	31	62	22	148
199	168	191	193	158	227	178	143	182	106	36	190
205	174	155	252	236	231	149	178	228	43	95	234
190	216	116	149	236	187	86	150	79	38	218	241
190	224	147	108	227	210	127	102	36	101	255	224
190	214	173	66	103	143	96	50	2	109	249	215
187	196	235	75	1	81	47	0	6	217	255	211
183	202	237	145	0	0	12	108	200	138	243	236
195	206	123	207	177	121	123	200	175	13	96	218

157	153	174	168	150	152	129	151	172	161	155	156
155	182	163	74	75	62	33	17	110	210	180	154
180	180	50	14	34	6	10	33	48	106	159	181
206	109	5	124	131	111	120	204	166	15	56	180
194	68	137	251	237	239	239	228	227	87	71	201
172	105	207	233	233	214	220	239	228	98	74	206
188	88	179	209	185	215	211	158	139	75	20	169
189	97	165	84	10	168	134	11	31	62	22	148
199	168	191	193	158	227	178	143	182	106	36	190
205	174	155	252	236	231	149	178	228	43	95	234
190	216	116	149	236	187	86	150	79	38	218	241
190	224	147	108	227	210	127	102	36	101	255	224
190	214	173	66	103	143	96	50	2	109	249	215
187	196	235	75	1	81	47	0	6	217	255	211
183	202	237	145	0	0	12	108	200	138	243	236
195	206	123	207	177	121	123	200	175	13	96	218

1 Unlocking the phone
2 Recognizing the images
3 Issue of normal neural network
4 CNN
5 Standard Classification of images
6 Types of Images
7 Colored image
8 Understanding the pixel grid
9 Edge Detection
10 Edge Detection importance
11 Detecting edges
12 Understanding Filter
13 Vertical Edge Detection
14 Horizontal Edge Detection
15 Convolution Operation
16 Kernel
17 Stride and Padding
18 Pooling Layer
19 CNN Architecture
20 Vertical Edge Detection
21 Detection
22 Resultant image
23 Filters
24 What Else Can Convolution Do?
25 How CNNs Use Layers
26 Applications
27 Training CNN
28 Famous CNN Architectures
29 CNN vs ANN vs RNN
30 HPC
Distributed memory
32 Data Parallelism
33 Model Parallelism
34 Pipeline Parallelism
35 Hybrid Parallelism

Standard Classification of images

Black and White Image

Grayscale Image

Colored Image

1 Unlocking the phone
2 Recognizing the images
3 Issue of normal neural network
4 CNN
5 Standard Classification of images
6 Types of Images
7 Colored image
8 Understanding the pixel grid
9 Edge Detection
10 Edge Detection importance
11 Detecting edges
12 Understanding Filter
13 Vertical Edge Detection
14 Horizontal Edge Detection
15 Convolution Operation
16 Kernel
17 Stride and Padding
18 Pooling Layer
19 CNN Architecture
20 Vertical Edge Detection
21 Detection
22 Resultant image
23 Filters
24 What Else Can Convolution Do?
25 How CNNs Use Layers
26 Applications
27 Training CNN
28 Famous CNN Architectures
29 CNN vs ANN vs RNN
30 HPC
Distributed memory
32 Data Parallelism
33 Model Parallelism
34 Pipeline Parallelism
35 Hybrid Parallelism

Types of Images Based on Pixel Representation

0	0	1	1
0	1	1	0
1	1	0	0
0	0	0	1

30	80	120	200
45	90	130	210
60	110	150	220
70	140	180	240

RED channel

255	0	0	255
255	128	0	255
0	100	50	200
10	30	90	180

Green channel

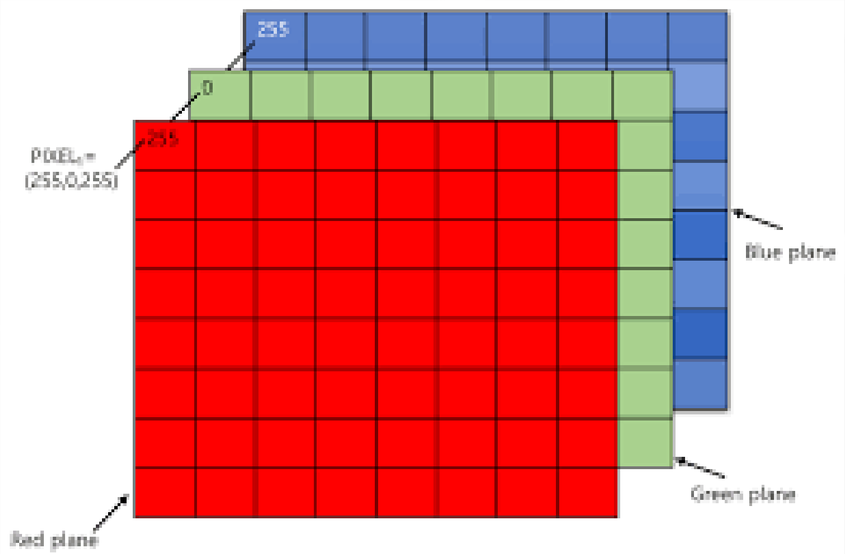
0	255	0	255
255	128	0	0
255	50	100	200
20	20	90	60

Blue channel

0	0	255	0
255	128	0	255
255	0	150	200
30	10	90	120

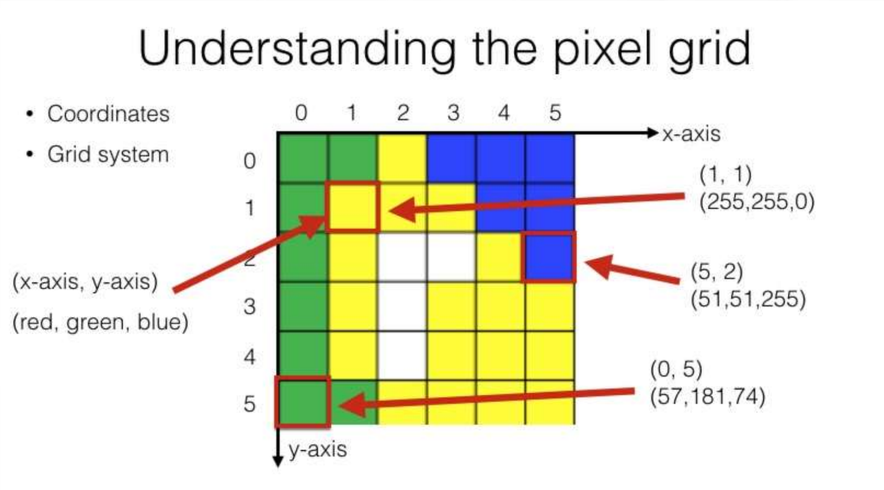
1 Unlocking the phone
2 Recognizing the images
3 Issue of normal neural network
4 CNN
5 Standard Classification of images
6 Types of Images
7 Colored image
8 Understanding the pixel grid
9 Edge Detection
10 Edge Detection importance
11 Detecting edges
12 Understanding Filter
13 Vertical Edge Detection
14 Horizontal Edge Detection
15 Convolution Operation
16 Kernel
17 Stride and Padding
18 Pooling Layer
19 CNN Architecture
20 Vertical Edge Detection
21 Detection
22 Resultant image
23 Filters
24 What Else Can Convolution Do?
25 How CNNs Use Layers
26 Applications
27 Training CNN
28 Famous CNN Architectures
29 CNN vs ANN vs RNN
30 HPC
Distributed memory
32 Data Parallelism
33 Model Parallelism
34 Pipeline Parallelism
35 Hybrid Parallelism

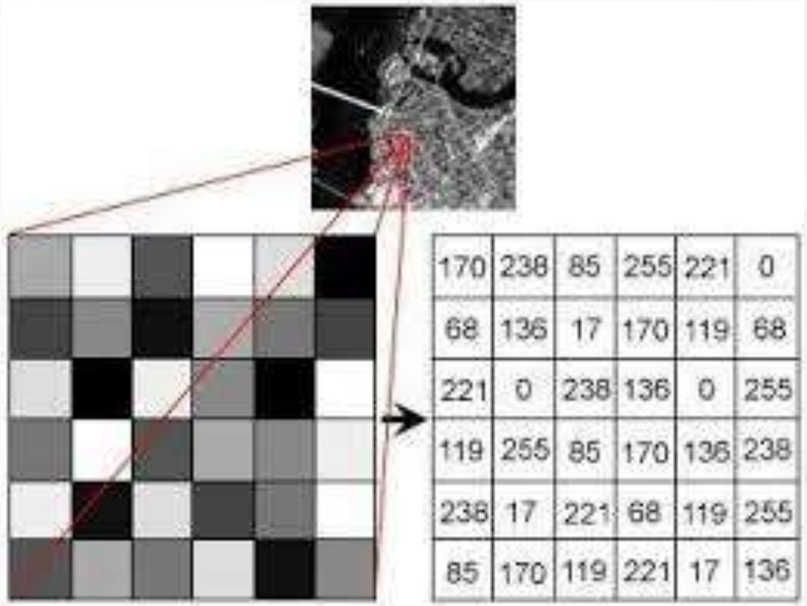
Colored image



1 Unlocking the phone
2 Recognizing the images
3 Issue of normal neural network
4 CNN
5 Standard Classification of images
6 Types of Images
7 Colored image
8 Understanding the pixel grid
9 Edge Detection
10 Edge Detection importance
11 Detecting edges
12 Understanding Filter
13 Vertical Edge Detection
14 Horizontal Edge Detection
15 Convolution Operation
16 Kernel
17 Stride and Padding
18 Pooling Layer
19 CNN Architecture
20 Vertical Edge Detection
21 Detection
22 Resultant image
23 Filters
24 What Else Can Convolution Do?
25 How CNNs Use Layers
26 Applications
27 Training CNN
28 Famous CNN Architectures
29 CNN vs ANN vs RNN
30 HPC
Distributed memory
32 Data Parallelism
33 Model Parallelism
34 Pipeline Parallelism
35 Hybrid Parallelism

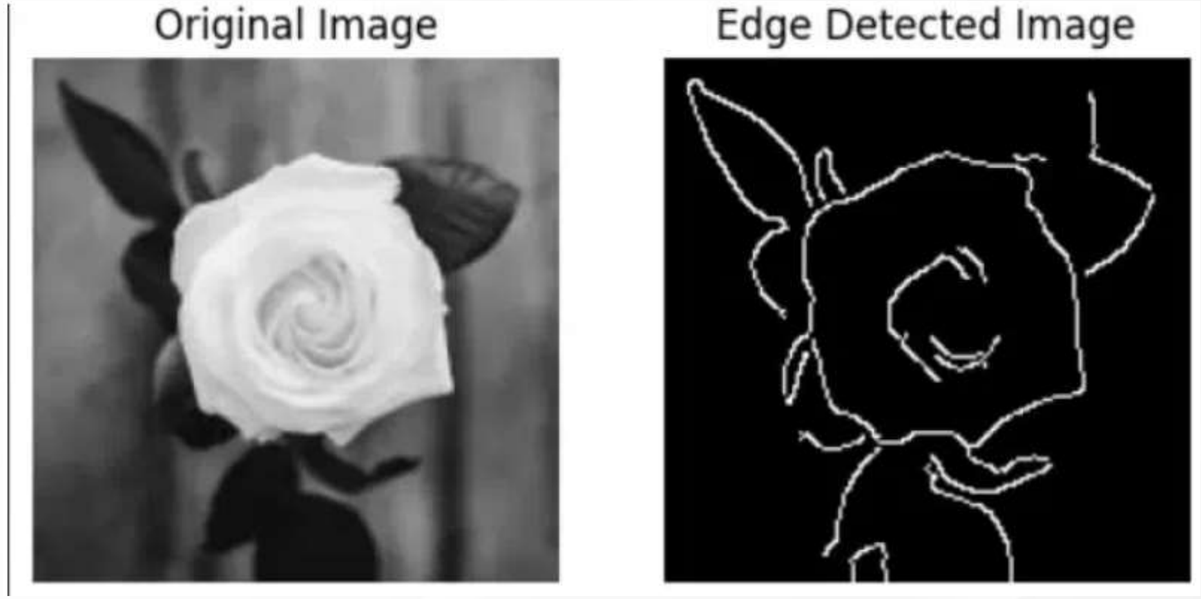
Understanding the pixel grid





1 Unlocking the phone
2 Recognizing the images
3 Issue of normal neural network
4 CNN
5 Standard Classification of images
6 Types of Images
7 Colored image
8 Understanding the pixel grid
9 Edge Detection
10 Edge Detection importance
11 Detecting edges
12 Understanding Filter
13 Vertical Edge Detection
14 Horizontal Edge Detection
15 Convolution Operation
16 Kernel
17 Stride and Padding
18 Pooling Layer
19 CNN Architecture
20 Vertical Edge Detection
21 Detection
22 Resultant image
23 Filters
24 What Else Can Convolution Do?
25 How CNNs Use Layers
26 Applications
27 Training CNN
28 Famous CNN Architectures
29 CNN vs ANN vs RNN
30 HPC
Distributed memory
32 Data Parallelism
33 Model Parallelism
34 Pipeline Parallelism
35 Hybrid Parallelism

Edge Detection



1 Unlocking the phone
2 Recognizing the images
3 Issue of normal neural network
4 CNN
5 Standard Classification of images
6 Types of Images
7 Colored image
8 Understanding the pixel grid
9 Edge Detection
10 Edge Detection importance
11 Detecting edges
12 Understanding Filter
13 Vertical Edge Detection
14 Horizontal Edge Detection
15 Convolution Operation
16 Kernel
17 Stride and Padding
18 Pooling Layer
19 CNN Architecture
20 Vertical Edge Detection
21 Detection
22 Resultant image
23 Filters
24 What Else Can Convolution Do?
25 How CNNs Use Layers
26 Applications
27 Training CNN
28 Famous CNN Architectures
29 CNN vs ANN vs RNN
30 HPC
Distributed memory
32 Data Parallelism
33 Model Parallelism
34 Pipeline Parallelism
35 Hybrid Parallelism

Edge Detection importance

It Turns "Data" into "Meaning"

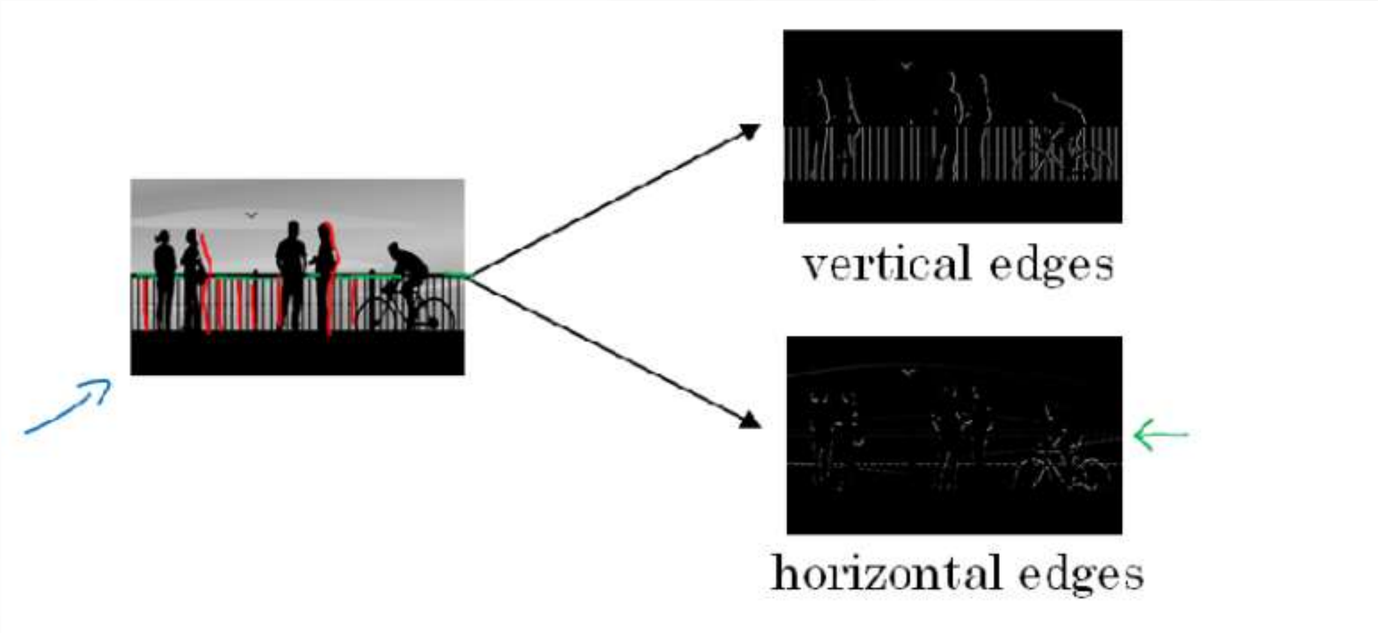
- Images are just huge grids of numbers (pixels). For a computer, seeing a million "50s" and "100s" is confusing. Edge detection acts like a **highlighter**: it throws away the useless background and keeps only the shape.

It Mimics How the Human Eye Works

- Our own brains are wired to notice contrast first. When you look at a flower, you don't process every single shade of pink; your eyes automatically find the **outline** of the petals.

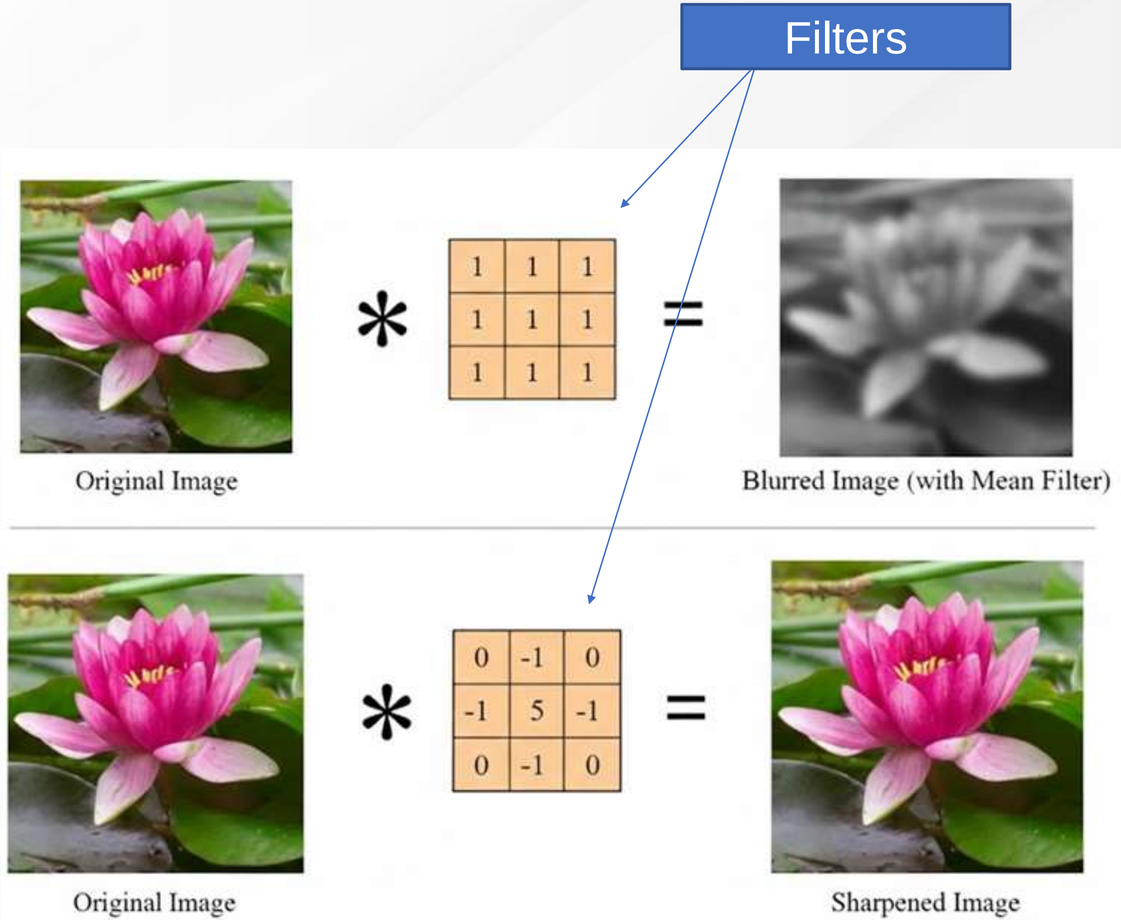
1 Unlocking the phone
2 Recognizing the images
3 Issue of normal neural network
4 CNN
5 Standard Classification of images
6 Types of Images
7 Colored image
8 Understanding the pixel grid
9 Edge Detection
10 Edge Detection importance
11 Detecting edges
12 Understanding Filter
13 Vertical Edge Detection
14 Horizontal Edge Detection
15 Convolution Operation
16 Kernel
17 Stride and Padding
18 Pooling Layer
19 CNN Architecture
20 Vertical Edge Detection
21 Detection
22 Resultant image
23 Filters
24 What Else Can Convolution Do?
25 How CNNs Use Layers
26 Applications
27 Training CNN
28 Famous CNN Architectures
29 CNN vs ANN vs RNN
30 HPC
Distributed memory
32 Data Parallelism
33 Model Parallelism
34 Pipeline Parallelism
35 Hybrid Parallelism

Detecting edges



1 Unlocking the phone
2 Recognizing the images
3 Issue of normal neural network
4 CNN
5 Standard Classification of images
6 Types of Images
7 Colored image
8 Understanding the pixel grid
9 Edge Detection
10 Edge Detection importance
11 Detecting edges
12 Understanding Filter
13 Vertical Edge Detection
14 Horizontal Edge Detection
15 Convolution Operation
16 Kernel
17 Stride and Padding
18 Pooling Layer
19 CNN Architecture
20 Vertical Edge Detection
21 Detection
22 Resultant image
23 Filters
24 What Else Can Convolution Do?
25 How CNNs Use Layers
26 Applications
27 Training CNN
28 Famous CNN Architectures
29 CNN vs ANN vs RNN
30 HPC
Distributed memory
32 Data Parallelism
33 Model Parallelism
34 Pipeline Parallelism
35 Hybrid Parallelism

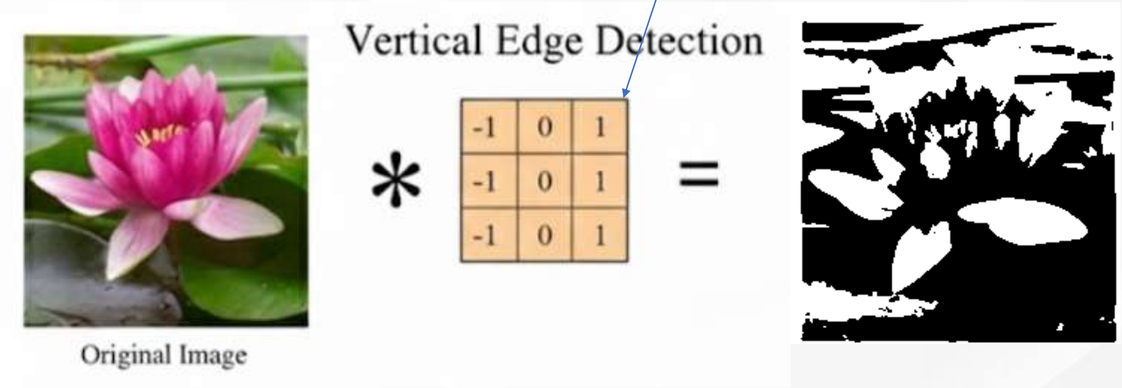
Understanding Filter



1 Unlocking the phone
2 Recognizing the images
3 Issue of normal neural network
4 CNN
5 Standard Classification of images
6 Types of Images
7 Colored image
8 Understanding the pixel grid
9 Edge Detection
10 Edge Detection importance
11 Detecting edges
12 Understanding Filter
13 Vertical Edge Detection
14 Horizontal Edge Detection
15 Convolution Operation
16 Kernel
17 Stride and Padding
18 Pooling Layer
19 CNN Architecture
20 Vertical Edge Detection
21 Detection
22 Resultant image
23 Filters
24 What Else Can Convolution Do?
25 How CNNs Use Layers
26 Applications
27 Training CNN
28 Famous CNN Architectures
29 CNN vs ANN vs RNN
30 HPC
Distributed memory
32 Data Parallelism
33 Model Parallelism
34 Pipeline Parallelism
35 Hybrid Parallelism

Understanding Filter

Filters



1 Unlocking the phone
2 Recognizing the images
3 Issue of normal neural network
4 CNN
5 Standard Classification of images
6 Types of Images
7 Colored image
8 Understanding the pixel grid
9 Edge Detection
10 Edge Detection importance
11 Detecting edges
12 Understanding Filter
13 Vertical Edge Detection
14 Horizontal Edge Detection
15 Convolution Operation
16 Kernel
17 Stride and Padding
18 Pooling Layer
19 CNN Architecture
20 Vertical Edge Detection
21 Detection
22 Resultant image
23 Filters
24 What Else Can Convolution Do?
25 How CNNs Use Layers
26 Applications
27 Training CNN
28 Famous CNN Architectures
29 CNN vs ANN vs RNN
30 HPC
Distributed memory
32 Data Parallelism
33 Model Parallelism
34 Pipeline Parallelism
35 Hybrid Parallelism

Vertical Edge Detection

1. Vertical Edge Detection

- **What it finds:** Vertical lines (like the side of a building, a tree trunk, or the stem of your flower).
- **How it works:** It looks for a change in brightness as you move horizontally across the image.
- **The Kernel:** The positive and negative numbers are in the columns.

$$\text{Kernel} = \begin{bmatrix} 1 & 0 & -1 \\ 1 & 0 & -1 \\ 1 & 0 & -1 \end{bmatrix}$$

- **Real-world logic:** If the left side is bright (1) and the right side is dark (0), the result is a high number, signaling a vertical boundary.

1 Unlocking the phone
2 Recognizing the images
3 Issue of normal neural network
4 CNN
5 Standard Classification of images
6 Types of Images
7 Colored image
8 Understanding the pixel grid
9 Edge Detection
10 Edge Detection importance
11 Detecting edges
12 Understanding Filter
13 Vertical Edge Detection
14 Horizontal Edge Detection
15 Convolution Operation
16 Kernel
17 Stride and Padding
18 Pooling Layer
19 CNN Architecture
20 Vertical Edge Detection
21 Detection
22 Resultant image
23 Filters
24 What Else Can Convolution Do?
25 How CNNs Use Layers
26 Applications
27 Training CNN
28 Famous CNN Architectures
29 CNN vs ANN vs RNN
30 HPC
Distributed memory
32 Data Parallelism
33 Model Parallelism
34 Pipeline Parallelism
35 Hybrid Parallelism

Horizontal Edge Detection

2. Horizontal Edge Detection

- **What it finds:** Horizontal lines (like the horizon, table edges, or stripes on a wall).
- **How it works:** It looks for a change in brightness as you move vertically across the image.
- **The Kernel:** The positive and negative numbers are in the rows.

$$\text{Kernel} = \begin{bmatrix} 1 & 1 & 1 \\ 0 & 0 & 0 \\ -1 & -1 & -1 \end{bmatrix}$$

- **Real-world logic:** If the top part is bright (1) and the bottom part is dark (-1), the result is a high number, signaling a horizontal boundary.

1 Unlocking the phone
2 Recognizing the images
3 Issue of normal neural network
4 CNN
5 Standard Classification of images
6 Types of Images
7 Colored image
8 Understanding the pixel grid
9 Edge Detection
10 Edge Detection importance
11 Detecting edges
12 Understanding Filter
13 Vertical Edge Detection
14 Horizontal Edge Detection
15 Convolution Operation
16 Kernel
17 Stride and Padding
18 Pooling Layer
19 CNN Architecture
20 Vertical Edge Detection
21 Detection
22 Resultant image
23 Filters
24 What Else Can Convolution Do?
25 How CNNs Use Layers
26 Applications
27 Training CNN
28 Famous CNN Architectures
29 CNN vs ANN vs RNN
30 HPC
Distributed memory
32 Data Parallelism
33 Model Parallelism
34 Pipeline Parallelism
35 Hybrid Parallelism

Convolution Operation

- **Convolution** is a mathematical operation that allows a CNN to **detect meaningful patterns** (edges, textures, shapes) from input data such as images.
- Magnifying Glass
- CNN slides a small window (filter/kernel) over an image.
- That window asks:
- “Do I see something interesting here?”

1 Unlocking the phone
2 Recognizing the images
3 Issue of normal neural network
4 CNN
5 Standard Classification of images
6 Types of Images
7 Colored image
8 Understanding the pixel grid
9 Edge Detection
10 Edge Detection importance
11 Detecting edges
12 Understanding Filter
13 Vertical Edge Detection
14 Horizontal Edge Detection
15 Convolution Operation
16 Kernel
17 Stride and Padding
18 Pooling Layer
19 CNN Architecture
20 Vertical Edge Detection
21 Detection
22 Resultant image
23 Filters
24 What Else Can Convolution Do?
25 How CNNs Use Layers
26 Applications
27 Training CNN
28 Famous CNN Architectures
29 CNN vs ANN vs RNN
30 HPC
Distributed memory
32 Data Parallelism
33 Model Parallelism
34 Pipeline Parallelism
35 Hybrid Parallelism

Kernel

Image (5x5)
Kernel (3x3)

Kernel slides → dot product → feature map

What Kernels Learn:
Edges
Corners
Lines
Textures
Each kernel = feature detector

Output = Σ(pixel × kernel) + bias

Multiply → add → activate

1 Unlocking the phone
2 Recognizing the images
3 Issue of normal neural network
4 CNN
5 Standard Classification of images
6 Types of Images
7 Colored image
8 Understanding the pixel grid
9 Edge Detection
10 Edge Detection importance
11 Detecting edges
12 Understanding Filter
13 Vertical Edge Detection
14 Horizontal Edge Detection
15 Convolution Operation
16 Kernel
17 Stride and Padding
18 Pooling Layer
19 CNN Architecture
20 Vertical Edge Detection
21 Detection
22 Resultant image
23 Filters
24 What Else Can Convolution Do?
25 How CNNs Use Layers
26 Applications
27 Training CNN
28 Famous CNN Architectures
29 CNN vs ANN vs RNN
30 HPC
Distributed memory
32 Data Parallelism
33 Model Parallelism
34 Pipeline Parallelism
35 Hybrid Parallelism

Stride and Padding

Stride & Padding (Intuition)

Stride:

How fast the kernel moves

Bigger stride → smaller output

Padding:

Add border zeros

Prevent size shrinking

Analogy:

Walking steps vs baby steps

1 Unlocking the phone
2 Recognizing the images
3 Issue of normal neural network
4 CNN
5 Standard Classification of images
6 Types of Images
7 Colored image
8 Understanding the pixel grid
9 Edge Detection
10 Edge Detection importance
11 Detecting edges
12 Understanding Filter
13 Vertical Edge Detection
14 Horizontal Edge Detection
15 Convolution Operation
16 Kernel
17 Stride and Padding
18 Pooling Layer
19 CNN Architecture
20 Vertical Edge Detection
21 Detection
22 Resultant image
23 Filters
24 What Else Can Convolution Do?
25 How CNNs Use Layers
26 Applications
27 Training CNN
28 Famous CNN Architectures
29 CNN vs ANN vs RNN
30 HPC
Distributed memory
32 Data Parallelism
33 Model Parallelism
34 Pipeline Parallelism
35 Hybrid Parallelism

Pooling Layer

Pooling = Information Compression
Max Pooling:
"Keep the strongest signal"
Why?
Reduce computation
Prevent overfitting
Keep important features

Reading a paragraph → remembering keywords only

1 Unlocking the phone
2 Recognizing the images
3 Issue of normal neural network
4 CNN
5 Standard Classification of images
6 Types of Images
7 Colored image
8 Understanding the pixel grid
9 Edge Detection
10 Edge Detection importance
11 Detecting edges
12 Understanding Filter
13 Vertical Edge Detection
14 Horizontal Edge Detection
15 Convolution Operation
16 Kernel
17 Stride and Padding
18 Pooling Layer
19 CNN Architecture
20 Vertical Edge Detection
21 Detection
22 Resultant image
23 Filters
24 What Else Can Convolution Do?
25 How CNNs Use Layers
26 Applications
27 Training CNN
28 Famous CNN Architectures
29 CNN vs ANN vs RNN
30 HPC
Distributed memory
32 Data Parallelism
33 Model Parallelism
34 Pipeline Parallelism
35 Hybrid Parallelism

CNN Architecture

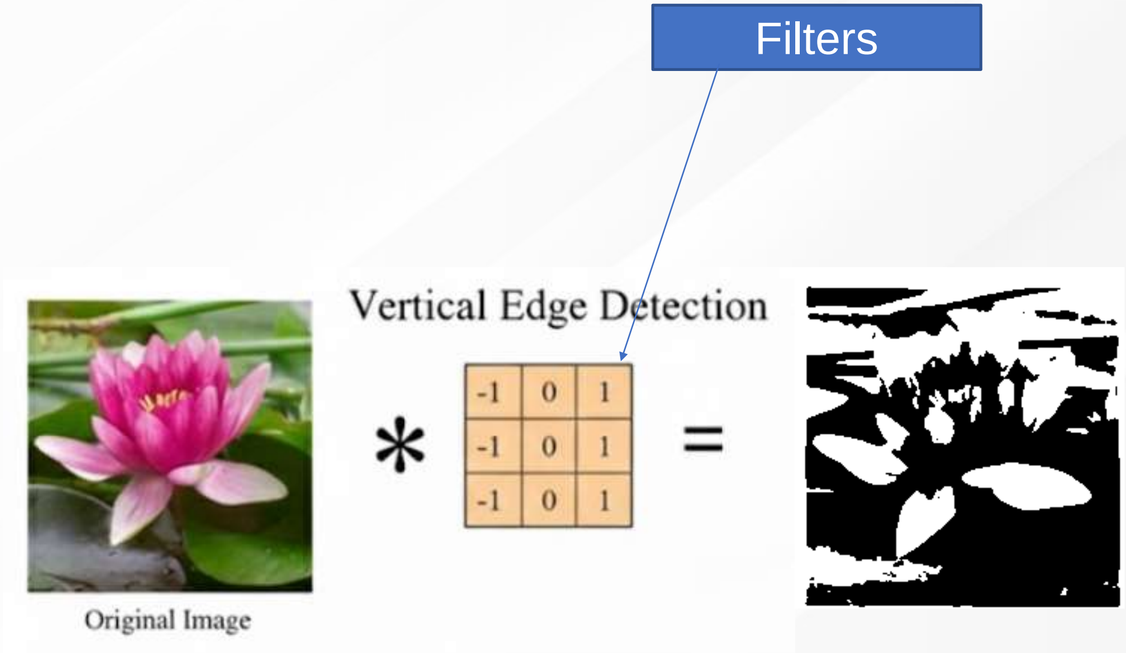
Image
↓
Convolution
↓
ReLU
↓
Pooling
↓
Convolution
↓
Pooling
↓
Flatten
↓
Dense
↓
Output

Why ReLU?
Without ReLU:
Network becomes linear
Learns nothing complex
ReLU:
$f(x) = \max(0, x)$

Flatten Layer
Turns:
Feature maps → 1D vector
For:
Classification
Regression

1 Unlocking the phone
2 Recognizing the images
3 Issue of normal neural network
4 CNN
5 Standard Classification of images
6 Types of Images
7 Colored image
8 Understanding the pixel grid
9 Edge Detection
10 Edge Detection importance
11 Detecting edges
12 Understanding Filter
13 Vertical Edge Detection
14 Horizontal Edge Detection
15 Convolution Operation
16 Kernel
17 Stride and Padding
18 Pooling Layer
19 CNN Architecture
20 Vertical Edge Detection
21 Detection
22 Resultant image
23 Filters
24 What Else Can Convolution Do?
25 How CNNs Use Layers
26 Applications
27 Training CNN
28 Famous CNN Architectures
29 CNN vs ANN vs RNN
30 HPC
Distributed memory
32 Data Parallelism
33 Model Parallelism
34 Pipeline Parallelism
35 Hybrid Parallelism

Vertical Edge Detection



1 Unlocking the phone
2 Recognizing the images
3 Issue of normal neural network
4 CNN
5 Standard Classification of images
6 Types of Images
7 Colored image
8 Understanding the pixel grid
9 Edge Detection
10 Edge Detection importance
11 Detecting edges
12 Understanding Filter
13 Vertical Edge Detection
14 Horizontal Edge Detection
15 Convolution Operation
16 Kernel
17 Stride and Padding
18 Pooling Layer
19 CNN Architecture
20 Vertical Edge Detection
21 Detection
22 Resultant image
23 Filters
24 What Else Can Convolution Do?
25 How CNNs Use Layers
26 Applications
27 Training CNN
28 Famous CNN Architectures
29 CNN vs ANN vs RNN
30 HPC
Distributed memory
32 Data Parallelism
33 Model Parallelism
34 Pipeline Parallelism
35 Hybrid Parallelism

Vertical Edge Detection

$$\begin{matrix} 1 & 1 & \mathbf{0} & 0 & 0 \\ 1 & 1 & \mathbf{0} & 0 & 0 \\ 1 & 1 & \mathbf{0} & 0 & 0 \\ 1 & 1 & \mathbf{0} & 0 & 0 \\ 1 & 1 & \mathbf{0} & 0 & 0 \end{matrix} * \begin{matrix} 1 & 0 & -1 \\ 1 & 1 & -1 \\ 1 & 1 & -1 \end{matrix}$$

1 Unlocking the phone
2 Recognizing the images
3 Issue of normal neural network
4 CNN
5 Standard Classification of images
6 Types of Images
7 Colored image
8 Understanding the pixel grid
9 Edge Detection
10 Edge Detection importance
11 Detecting edges
12 Understanding Filter
13 Vertical Edge Detection
14 Horizontal Edge Detection
15 Convolution Operation
16 Kernel
17 Stride and Padding
18 Pooling Layer
19 CNN Architecture
20 Vertical Edge Detection
21 Detection
22 Resultant image
23 Filters
24 What Else Can Convolution Do?
25 How CNNs Use Layers
26 Applications
27 Training CNN
28 Famous CNN Architectures
29 CNN vs ANN vs RNN
30 HPC
Distributed memory
32 Data Parallelism
33 Model Parallelism
34 Pipeline Parallelism
35 Hybrid Parallelism

$$\begin{bmatrix} 1 & 1 & \mathbf{0} & 0 & 0 \\ 1 & 1 & \mathbf{0} & 0 & 0 \\ 1 & 1 & \mathbf{0} & 0 & 0 \\ 1 & 1 & \mathbf{0} & 0 & 0 \\ 1 & 1 & \mathbf{0} & 0 & 0 \end{bmatrix} * \begin{bmatrix} 1 & 0 & -1 \\ 1 & 1 & -1 \\ 1 & 1 & -1 \end{bmatrix}$$

Top Row: (1 × 1) + (1 × 0) + (0 × -1) = **1**
Middle Row: (1 × 1) + (1 × 1) + (0 × -1) = **2**
Bottom Row: (1 × 1) + (1 × 1) + (0 × -1) = **2**
ie 1+2+2=5

1 Unlocking the phone
2 Recognizing the images
3 Issue of normal neural network
4 CNN
5 Standard Classification of images
6 Types of Images
7 Colored image
8 Understanding the pixel grid
9 Edge Detection
10 Edge Detection importance
11 Detecting edges
12 Understanding Filter
13 Vertical Edge Detection
14 Horizontal Edge Detection
15 Convolution Operation
16 Kernel
17 Stride and Padding
18 Pooling Layer
19 CNN Architecture
20 Vertical Edge Detection
21 Detection
22 Resultant image
23 Filters
24 What Else Can Convolution Do?
25 How CNNs Use Layers
26 Applications
27 Training CNN
28 Famous CNN Architectures
29 CNN vs ANN vs RNN
30 HPC
Distributed memory
32 Data Parallelism
33 Model Parallelism
34 Pipeline Parallelism
35 Hybrid Parallelism

$$\begin{bmatrix} 1 & 1 & \mathbf{0} & 0 & 0 \\ 1 & 1 & \mathbf{0} & 0 & 0 \\ 1 & 1 & \mathbf{0} & 0 & 0 \\ 1 & 1 & \mathbf{0} & 0 & 0 \\ 1 & 1 & \mathbf{0} & 0 & 0 \end{bmatrix} \overset{*}{*} \begin{bmatrix} 1 & 0 & -1 \\ 1 & 1 & -1 \\ 1 & 1 & -1 \end{bmatrix} = \begin{bmatrix} \boxed{5} & ? & ? \\ ? & ? & ? \\ ? & ? & ? \end{bmatrix}$$

But why 3*3

1 Unlocking the phone
2 Recognizing the images
3 Issue of normal neural network
4 CNN
5 Standard Classification of images
6 Types of Images
7 Colored image
8 Understanding the pixel grid
9 Edge Detection
10 Edge Detection importance
11 Detecting edges
12 Understanding Filter
13 Vertical Edge Detection
14 Horizontal Edge Detection
15 Convolution Operation
16 Kernel
17 Stride and Padding
18 Pooling Layer
19 CNN Architecture
20 Vertical Edge Detection
21 Detection
22 Resultant image
23 Filters
24 What Else Can Convolution Do?
25 How CNNs Use Layers
26 Applications
27 Training CNN
28 Famous CNN Architectures
29 CNN vs ANN vs RNN
30 HPC
Distributed memory
32 Data Parallelism
33 Model Parallelism
34 Pipeline Parallelism
35 Hybrid Parallelism

$$
\begin{bmatrix} 1 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 0 & 0 \end{bmatrix} * \begin{bmatrix} 1 & 0 & -1 \\ 1 & 1 & -1 \\ 1 & 1 & -1 \end{bmatrix} = \begin{bmatrix} 5 & \boxed{?} & ? \\ ? & ? & ? \\ ? & ? & ? \end{bmatrix}
$$

But why 3*3

The 3×3 filter slides across the image one pixel at a time, moving through exactly 3 horizontal positions and 3 vertical positions before hitting the edges. Each of these 9 unique landing spots produces one calculation, perfectly filling out the 3×3 final matrix.

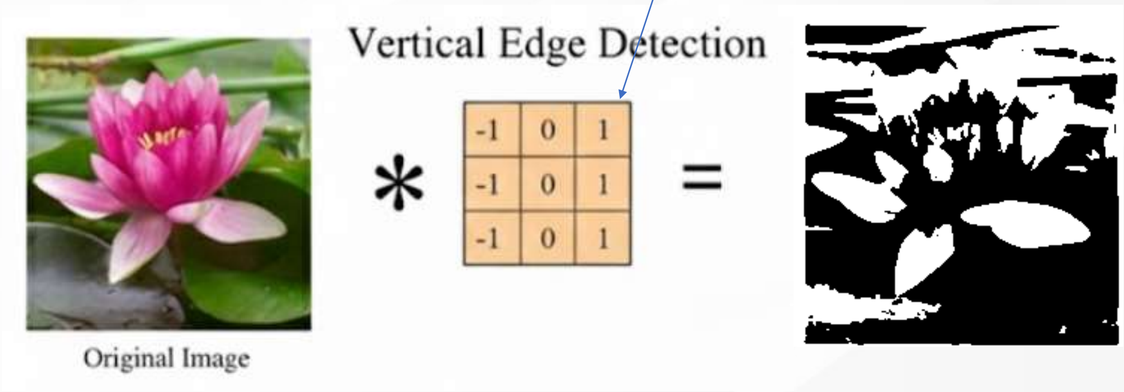
1 Unlocking the phone
2 Recognizing the images
3 Issue of normal neural network
4 CNN
5 Standard Classification of images
6 Types of Images
7 Colored image
8 Understanding the pixel grid
9 Edge Detection
10 Edge Detection importance
11 Detecting edges
12 Understanding Filter
13 Vertical Edge Detection
14 Horizontal Edge Detection
15 Convolution Operation
16 Kernel
17 Stride and Padding
18 Pooling Layer
19 CNN Architecture
20 Vertical Edge Detection
21 Detection
22 Resultant image
23 Filters
24 What Else Can Convolution Do?
25 How CNNs Use Layers
26 Applications
27 Training CNN
28 Famous CNN Architectures
29 CNN vs ANN vs RNN
30 HPC
Distributed memory
32 Data Parallelism
33 Model Parallelism
34 Pipeline Parallelism
35 Hybrid Parallelism

$$\begin{bmatrix} 1 & 1 & \mathbf{0} & 0 & 0 \\ 1 & 1 & \mathbf{0} & 0 & 0 \\ 1 & 1 & \mathbf{0} & 0 & 0 \\ 1 & 1 & \mathbf{0} & 0 & 0 \\ 1 & 1 & \mathbf{0} & 0 & 0 \end{bmatrix} * \begin{bmatrix} 1 & 0 & -1 \\ 1 & 1 & -1 \\ 1 & 1 & -1 \end{bmatrix} = \begin{bmatrix} 5 & 0 & 1 \\ 4 & 2 & 2 \\ 3 & 1 & 0 \end{bmatrix}$$

resultant image after edge detection

1 Unlocking the phone
2 Recognizing the images
3 Issue of normal neural network
4 CNN
5 Standard Classification of images
6 Types of Images
7 Colored image
8 Understanding the pixel grid
9 Edge Detection
10 Edge Detection importance
11 Detecting edges
12 Understanding Filter
13 Vertical Edge Detection
14 Horizontal Edge Detection
15 Convolution Operation
16 Kernel
17 Stride and Padding
18 Pooling Layer
19 CNN Architecture
20 Vertical Edge Detection
21 Detection
22 Resultant image
23 Filters
24 What Else Can Convolution Do?
25 How CNNs Use Layers
26 Applications
27 Training CNN
28 Famous CNN Architectures
29 CNN vs ANN vs RNN
30 HPC
Distributed memory
32 Data Parallelism
33 Model Parallelism
34 Pipeline Parallelism
35 Hybrid Parallelism

Filters



1 Unlocking the phone
2 Recognizing the images
3 Issue of normal neural network
4 CNN
5 Standard Classification of images
6 Types of Images
7 Colored image
8 Understanding the pixel grid
9 Edge Detection
10 Edge Detection importance
11 Detecting edges
12 Understanding Filter
13 Vertical Edge Detection
14 Horizontal Edge Detection
15 Convolution Operation
16 Kernel
17 Stride and Padding
18 Pooling Layer
19 CNN Architecture
20 Vertical Edge Detection
21 Detection
22 Resultant image
23 Filters
24 What Else Can Convolution Do?
25 How CNNs Use Layers
26 Applications
27 Training CNN
28 Famous CNN Architectures
29 CNN vs ANN vs RNN
30 HPC
Distributed memory
32 Data Parallelism
33 Model Parallelism
34 Pipeline Parallelism
35 Hybrid Parallelism

What Else Can Convolution Do?

Beyond finding edges, here are the three most common real-world uses:

1. Blurring (Smoothing)

2. Sharpening

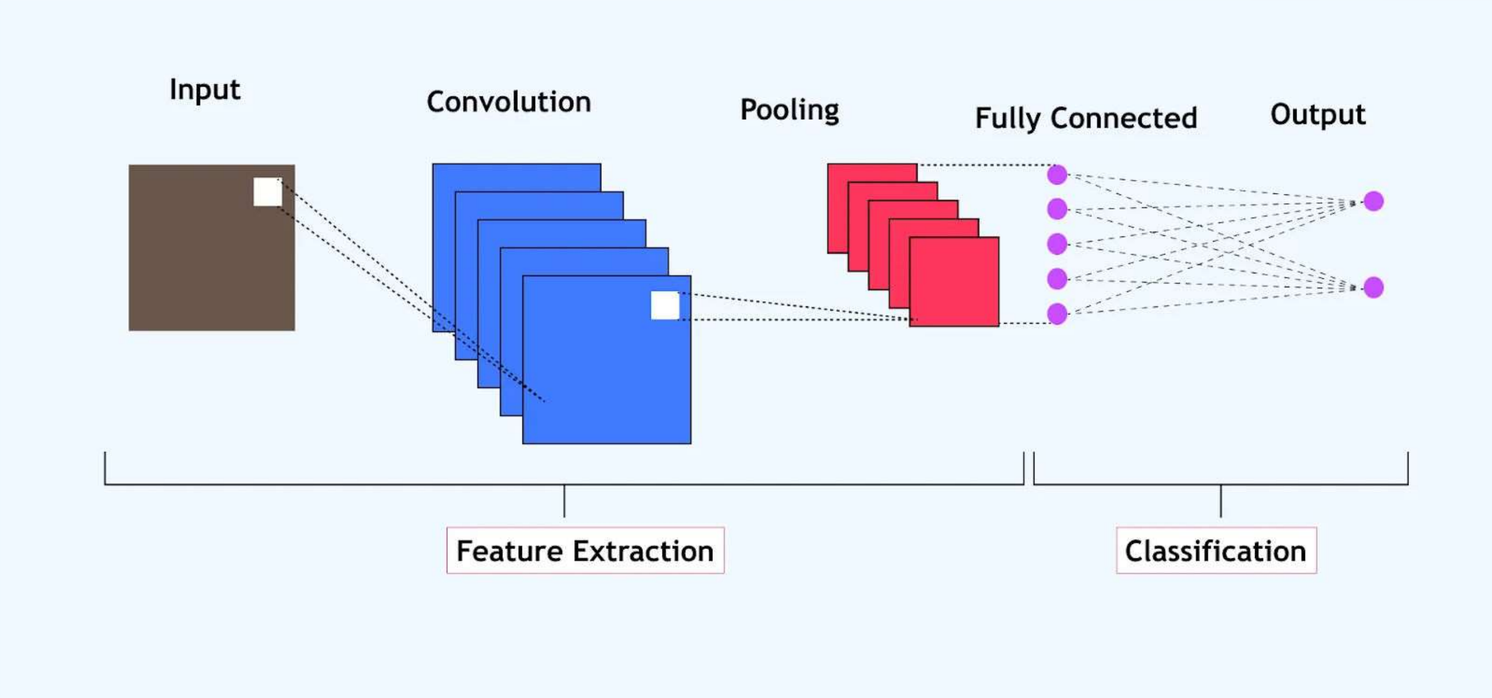
3. Feature Extraction (Deep Learning)

In Artificial Intelligence (like Face ID or self-driving cars), convolution isn't just looking for lines. It looks for **shapes**:

- **Layer 1:** Detects simple edges (vertical/horizontal).
- **Layer 2:** Detects textures and circles (using the results of Layer 1).
- **Layer 3:** Detects complex parts like eyes, noses, or wheels.

1 Unlocking the phone
2 Recognizing the images
3 Issue of normal neural network
4 CNN
5 Standard Classification of images
6 Types of Images
7 Colored image
8 Understanding the pixel grid
9 Edge Detection
10 Edge Detection importance
11 Detecting edges
12 Understanding Filter
13 Vertical Edge Detection
14 Horizontal Edge Detection
15 Convolution Operation
16 Kernel
17 Stride and Padding
18 Pooling Layer
19 CNN Architecture
20 Vertical Edge Detection
21 Detection
22 Resultant image
23 Filters
24 What Else Can Convolution Do?
25 How CNNs Use Layers
26 Applications
27 Training CNN
28 Famous CNN Architectures
29 CNN vs ANN vs RNN
30 HPC
Distributed memory
32 Data Parallelism
33 Model Parallelism
34 Pipeline Parallelism
35 Hybrid Parallelism

CNN



1 Unlocking the phone
2 Recognizing the images
3 Issue of normal neural network
4 CNN
5 Standard Classification of images
6 Types of Images
7 Colored image
8 Understanding the pixel grid
9 Edge Detection
10 Edge Detection importance
11 Detecting edges
12 Understanding Filter
13 Vertical Edge Detection
14 Horizontal Edge Detection
15 Convolution Operation
16 Kernel
17 Stride and Padding
18 Pooling Layer
19 CNN Architecture
20 Vertical Edge Detection
21 Detection
22 Resultant image
23 Filters
24 What Else Can Convolution Do?
25 How CNNs Use Layers
26 Applications
27 Training CNN
28 Famous CNN Architectures
29 CNN vs ANN vs RNN
30 HPC
Distributed memory
32 Data Parallelism
33 Model Parallelism
34 Pipeline Parallelism
35 Hybrid Parallelism

CNN

- Let us understand face recognition
- Imagine we are training **a** model for **person recognition** in images.

1 Unlocking the phone
2 Recognizing the images
3 Issue of normal neural network
4 CNN
5 Standard Classification of images
6 Types of Images
7 Colored image
8 Understanding the pixel grid
9 Edge Detection
10 Edge Detection importance
11 Detecting edges
12 Understanding Filter
13 Vertical Edge Detection
14 Horizontal Edge Detection
15 Convolution Operation
16 Kernel
17 Stride and Padding
18 Pooling Layer
19 CNN Architecture
20 Vertical Edge Detection
21 Detection
22 Resultant image
23 Filters
24 What Else Can Convolution Do?
25 How CNNs Use Layers
26 Applications
27 Training CNN
28 Famous CNN Architectures
29 CNN vs ANN vs RNN
30 HPC
Distributed memory
32 Data Parallelism
33 Model Parallelism
34 Pipeline Parallelism
35 Hybrid Parallelism

How CNNs Use Convolution, Pooling, and Fully Connected Layers)

Stage 1: Convolution (The Feature Finder)

Convolution is like a **magnifying glass** that scans the image to find specific "clues" or features.

- **Layer 1 (The Sketcher):** It looks for the simplest things—tiny vertical and horizontal edges (like the line of a jaw or the bridge of a nose).
- **Layer 2 (The Shape Maker):** It combines those lines to find "textures" and simple shapes like the circle of an eye or the curve of a nostril.
- **Layer 3 (The Object Builder):** It puts those shapes together to recognize complex parts, like an entire ear, a mouth, or a pair of glasses.

1 Unlocking the phone
2 Recognizing the images
3 Issue of normal neural network
4 CNN
5 Standard Classification of images
6 Types of Images
7 Colored image
8 Understanding the pixel grid
9 Edge Detection
10 Edge Detection importance
11 Detecting edges
12 Understanding Filter
13 Vertical Edge Detection
14 Horizontal Edge Detection
15 Convolution Operation
16 Kernel
17 Stride and Padding
18 Pooling Layer
19 CNN Architecture
20 Vertical Edge Detection
21 Detection
22 Resultant image
23 Filters
24 What Else Can Convolution Do?
25 How CNNs Use Layers
26 Applications
27 Training CNN
28 Famous CNN Architectures
29 CNN vs ANN vs RNN
30 HPC
Distributed memory
32 Data Parallelism
33 Model Parallelism
34 Pipeline Parallelism
35 Hybrid Parallelism

Stage 2: Pooling (The Summarizer)

After finding all those features, the computer has too much data to handle. **Pooling** simplifies the map while keeping the "best" parts.

- Imagine looking at a 1,000-word essay and only circling the **bold** words. You get the meaning much faster.
- It makes the AI "flexible." It tells the computer: *"I don't care exactly which pixel the nose is on; I just need to know there IS a nose in the middle of the face"*.
- It shrinks the image size, making the computer much faster.

1 Unlocking the phone
2 Recognizing the images
3 Issue of normal neural network
4 CNN
5 Standard Classification of images
6 Types of Images
7 Colored image
8 Understanding the pixel grid
9 Edge Detection
10 Edge Detection importance
11 Detecting edges
12 Understanding Filter
13 Vertical Edge Detection
14 Horizontal Edge Detection
15 Convolution Operation
16 Kernel
17 Stride and Padding
18 Pooling Layer
19 CNN Architecture
20 Vertical Edge Detection
21 Detection
22 Resultant image
23 Filters
24 What Else Can Convolution Do?
25 How CNNs Use Layers
26 Applications
27 Training CNN
28 Famous CNN Architectures
29 CNN vs ANN vs RNN
30 HPC
Distributed memory
32 Data Parallelism
33 Model Parallelism
34 Pipeline Parallelism
35 Hybrid Parallelism

Stage 3: Fully Connected Layer (The Judge)

- This is the final "thinking" step where all the clues are put together to make a final decision.
- This layer looks at all the summarized clues from the previous steps (e.g., 2 Eyes + 1 Nose + 1 Mouth).
- It connects these clues to its memory of different people. It says: *"Based on these specific features, there is a 99% chance this is Ram"*.
- It is the **Judge** who looks at all the evidence and gives the final answer.

1 Unlocking the phone
2 Recognizing the images
3 Issue of normal neural network
4 CNN
5 Standard Classification of images
6 Types of Images
7 Colored image
8 Understanding the pixel grid
9 Edge Detection
10 Edge Detection importance
11 Detecting edges
12 Understanding Filter
13 Vertical Edge Detection
14 Horizontal Edge Detection
15 Convolution Operation
16 Kernel
17 Stride and Padding
18 Pooling Layer
19 CNN Architecture
20 Vertical Edge Detection
21 Detection
22 Resultant image
23 Filters
24 What Else Can Convolution Do?
25 How CNNs Use Layers
26 Applications
27 Training CNN
28 Famous CNN Architectures
29 CNN vs ANN vs RNN
30 HPC
Distributed memory
32 Data Parallelism
33 Model Parallelism
34 Pipeline Parallelism
35 Hybrid Parallelism

Applications

- **Facial Recognition:** Unlocking phones (FaceID) and tagging friends on social media.
- **Self-Driving Cars:** Identifying pedestrians, traffic lights, and lane markings.
- **Medical Imaging:** Detecting tumors, fractures, and diseases in X-rays or MRIs.
- **Object Detection:** Identifying products in retail (Google Lens) or scanning barcodes.
- **Security & Surveillance:** Detecting intruders or abandoned baggage in public spaces.
- **Satellite Imagery:** Tracking deforestation, urban growth, and weather patterns.
- **Handwriting Recognition:** Converting handwritten notes or mail addresses into digital text.

1 Unlocking the phone
2 Recognizing the images
3 Issue of normal neural network
4 CNN
5 Standard Classification of images
6 Types of Images
7 Colored image
8 Understanding the pixel grid
9 Edge Detection
10 Edge Detection importance
11 Detecting edges
12 Understanding Filter
13 Vertical Edge Detection
14 Horizontal Edge Detection
15 Convolution Operation
16 Kernel
17 Stride and Padding
18 Pooling Layer
19 CNN Architecture
20 Vertical Edge Detection
21 Detection
22 Resultant image
23 Filters
24 What Else Can Convolution Do?
25 How CNNs Use Layers
26 Applications
27 Training CNN
28 Famous CNN Architectures
29 CNN vs ANN vs RNN
30 HPC
Distributed memory
32 Data Parallelism
33 Model Parallelism
34 Pipeline Parallelism
35 Hybrid Parallelism

Training CNN

CNN Learns:
Kernel values
Weights
Biases

Using:
Loss function
Backpropagation
Gradient descent

Insight:
CNN does NOT know what an eye or wheel is
It discovers them automatically

1 Unlocking the phone
2 Recognizing the images
3 Issue of normal neural network
4 CNN
5 Standard Classification of images
6 Types of Images
7 Colored image
8 Understanding the pixel grid
9 Edge Detection
10 Edge Detection importance
11 Detecting edges
12 Understanding Filter
13 Vertical Edge Detection
14 Horizontal Edge Detection
15 Convolution Operation
16 Kernel
17 Stride and Padding
18 Pooling Layer
19 CNN Architecture
20 Vertical Edge Detection
21 Detection
22 Resultant image
23 Filters
24 What Else Can Convolution Do?
25 How CNNs Use Layers
26 Applications
27 Training CNN
28 Famous CNN Architectures
29 CNN vs ANN vs RNN
30 HPC
Distributed memory
32 Data Parallelism
33 Model Parallelism
34 Pipeline Parallelism
35 Hybrid Parallelism

Famous CNN Architectures

Model	*and Contribution*
LeNet	First CNN
AlexNet	Deep CNN success
VGG	Very deep
ResNet	Skip connections
MobileNet	Lightweight

1 Unlocking the phone
2 Recognizing the images
3 Issue of normal neural network
4 CNN
5 Standard Classification of images
6 Types of Images
7 Colored image
8 Understanding the pixel grid
9 Edge Detection
10 Edge Detection importance
11 Detecting edges
12 Understanding Filter
13 Vertical Edge Detection
14 Horizontal Edge Detection
15 Convolution Operation
16 Kernel
17 Stride and Padding
18 Pooling Layer
19 CNN Architecture
20 Vertical Edge Detection
21 Detection
22 Resultant image
23 Filters
24 What Else Can Convolution Do?
25 How CNNs Use Layers
26 Applications
27 Training CNN
28 Famous CNN Architectures
29 CNN vs ANN vs RNN
30 HPC
Distributed memory
32 Data Parallelism
33 Model Parallelism
34 Pipeline Parallelism
35 Hybrid Parallelism

Applications

Computer Vision
Face recognition
Object detection
OCR

Healthcare
X-ray analysis
Tumor detection

Autonomous Cars
Lane detection
Traffic sign recognition

Cybersecurity
Malware image detection
Visual CAPTCHA breaking
Surveillance analysis

1 Unlocking the phone
2 Recognizing the images
3 Issue of normal neural network
4 CNN
5 Standard Classification of images
6 Types of Images
7 Colored image
8 Understanding the pixel grid
9 Edge Detection
10 Edge Detection importance
11 Detecting edges
12 Understanding Filter
13 Vertical Edge Detection
14 Horizontal Edge Detection
15 Convolution Operation
16 Kernel
17 Stride and Padding
18 Pooling Layer
19 CNN Architecture
20 Vertical Edge Detection
21 Detection
22 Resultant image
23 Filters
24 What Else Can Convolution Do?
25 How CNNs Use Layers
26 Applications
27 Training CNN
28 Famous CNN Architectures
29 CNN vs ANN vs RNN
30 HPC
Distributed memory
32 Data Parallelism
33 Model Parallelism
34 Pipeline Parallelism
35 Hybrid Parallelism

CNN vs ANN vs RNN

Model	Best For
ANN	Tabular data
CNN	Images & videos
RNN/LSTM	Sequences

1 Unlocking the phone
2 Recognizing the images
3 Issue of normal neural network
4 CNN
5 Standard Classification of images
6 Types of Images
7 Colored image
8 Understanding the pixel grid
9 Edge Detection
10 Edge Detection importance
11 Detecting edges
12 Understanding Filter
13 Vertical Edge Detection
14 Horizontal Edge Detection
15 Convolution Operation
16 Kernel
17 Stride and Padding
18 Pooling Layer
19 CNN Architecture
20 Vertical Edge Detection
21 Detection
22 Resultant image
23 Filters
24 What Else Can Convolution Do?
25 How CNNs Use Layers
26 Applications
27 Training CNN
28 Famous CNN Architectures
29 CNN vs ANN vs RNN
30 HPC
Distributed memory
32 Data Parallelism
33 Model Parallelism
34 Pipeline Parallelism
35 Hybrid Parallelism

HPC

AI without power is just theory

Can you train ChatGPT on your laptop?

HPC = Many computers working together
to solve **one big problem fast**.

Why AI Needs HPC
- Huge datasets
- Billions of parameters
- Complex simulations

Without HPC:
- Training takes years
- Models fail to scale

1 Unlocking the phone
2 Recognizing the images
3 Issue of normal neural network
4 CNN
5 Standard Classification of images
6 Types of Images
7 Colored image
8 Understanding the pixel grid
9 Edge Detection
10 Edge Detection importance
11 Detecting edges
12 Understanding Filter
13 Vertical Edge Detection
14 Horizontal Edge Detection
15 Convolution Operation
16 Kernel
17 Stride and Padding
18 Pooling Layer
19 CNN Architecture
20 Vertical Edge Detection
21 Detection
22 Resultant image
23 Filters
24 What Else Can Convolution Do?
25 How CNNs Use Layers
26 Applications
27 Training CNN
28 Famous CNN Architectures
29 CNN vs ANN vs RNN
30 HPC
Distributed memory
32 Data Parallelism
33 Model Parallelism
34 Pipeline Parallelism
35 Hybrid Parallelism

HPC Components:
CPUs (general computation)
GPUs (parallel math)
TPUs (AI-specific)
Clusters
Distributed memory

Parallelism Types:
Data parallelism
Model parallelism
Pipeline parallelism

Real Use Cases
Climate modeling
Genomics
Large language models
Satellite image analysis
Smart city simulations

1 Unlocking the phone
2 Recognizing the images
3 Issue of normal neural network
4 CNN
5 Standard Classification of images
6 Types of Images
7 Colored image
8 Understanding the pixel grid
9 Edge Detection
10 Edge Detection importance
11 Detecting edges
12 Understanding Filter
13 Vertical Edge Detection
14 Horizontal Edge Detection
15 Convolution Operation
16 Kernel
17 Stride and Padding
18 Pooling Layer
19 CNN Architecture
20 Vertical Edge Detection
21 Detection
22 Resultant image
23 Filters
24 What Else Can Convolution Do?
25 How CNNs Use Layers
26 Applications
27 Training CNN
28 Famous CNN Architectures
29 CNN vs ANN vs RNN
30 HPC
Distributed memory
32 Data Parallelism
33 Model Parallelism
34 Pipeline Parallelism
35 Hybrid Parallelism

Data Parallelism

"Same brain, different notebooks."

50 students
Same question paper
Each student solves different pages

Then all answers are combined.

That's data parallelism

GPU 1 → Model → Data chunk 1
GPU 2 → Model → Data chunk 2
GPU 3 → Model → Data chunk 3
GPU 4 → Model → Data chunk 4
↓
Gradient Aggregation

1 neural network
1 million images
4 GPUs
Each GPU:
Gets 250,000 images
Trains same model
Gradients are averaged

CNN image training
LSTM language models
Most common in practice

Model same, data split

1 Unlocking the phone
2 Recognizing the images
3 Issue of normal neural network
4 CNN
5 Standard Classification of images
6 Types of Images
7 Colored image
8 Understanding the pixel grid
9 Edge Detection
10 Edge Detection importance
11 Detecting edges
12 Understanding Filter
13 Vertical Edge Detection
14 Horizontal Edge Detection
15 Convolution Operation
16 Kernel
17 Stride and Padding
18 Pooling Layer
19 CNN Architecture
20 Vertical Edge Detection
21 Detection
22 Resultant image
23 Filters
24 What Else Can Convolution Do?
25 How CNNs Use Layers
26 Applications
27 Training CNN
28 Famous CNN Architectures
29 CNN vs ANN vs RNN
30 HPC
Distributed memory
32 Data Parallelism
33 Model Parallelism
34 Pipeline Parallelism
35 Hybrid Parallelism

Model Parallelism

One big brain, split across people

A textbook is too big:
Student A reads Chapter 1–3
Student B reads Chapter 4–6
Student C reads Chapter 7–10

GPU 1 → Layers 1–4
GPU 2 → Layers 5–8
GPU 3 → Layers 9–12

No one student holds the full book.

Forward pass flows **across GPUs**.

That's model parallelism.

Large Transformers (GPT, BERT)
Vision-language models
LLM training

Model split, data same

Neural network too large for 1 GPU
Layers split across GPUs

1 Unlocking the phone
2 Recognizing the images
3 Issue of normal neural network
4 CNN
5 Standard Classification of images
6 Types of Images
7 Colored image
8 Understanding the pixel grid
9 Edge Detection
10 Edge Detection importance
11 Detecting edges
12 Understanding Filter
13 Vertical Edge Detection
14 Horizontal Edge Detection
15 Convolution Operation
16 Kernel
17 Stride and Padding
18 Pooling Layer
19 CNN Architecture
20 Vertical Edge Detection
21 Detection
22 Resultant image
23 Filters
24 What Else Can Convolution Do?
25 How CNNs Use Layers
26 Applications
27 Training CNN
28 Famous CNN Architectures
29 CNN vs ANN vs RNN
30 HPC
Distributed memory
32 Data Parallelism
33 Model Parallelism
34 Pipeline Parallelism
35 Hybrid Parallelism

Pipeline Parallelism

"Assembly line learning"

Cooking in a restaurant:
Chef 1 → chopping
Chef 2 → cooking
Chef 3 → plating
Multiple dishes move simultaneously.

That's **pipeline parallelism**

Model split + batch streaming

Batch 1 → GPU 1 → GPU 2 → GPU 3
Batch 2 → GPU 1 → GPU 2 → GPU 3
Batch 3 → GPU 1 → GPU 2 → GPU 3

Usage:
Deep CNNs
Large Transformers
When both model and data are large

Model split into stages
Mini-batches flow through stages

1 Unlocking the phone
2 Recognizing the images
3 Issue of normal neural network
4 CNN
5 Standard Classification of images
6 Types of Images
7 Colored image
8 Understanding the pixel grid
9 Edge Detection
10 Edge Detection importance
11 Detecting edges
12 Understanding Filter
13 Vertical Edge Detection
14 Horizontal Edge Detection
15 Convolution Operation
16 Kernel
17 Stride and Padding
18 Pooling Layer
19 CNN Architecture
20 Vertical Edge Detection
21 Detection
22 Resultant image
23 Filters
24 What Else Can Convolution Do?
25 How CNNs Use Layers
26 Applications
27 Training CNN
28 Famous CNN Architectures
29 CNN vs ANN vs RNN
30 HPC
Distributed memory
32 Data Parallelism
33 Model Parallelism
34 Pipeline Parallelism
35 Hybrid Parallelism

Hybrid Parallelism

Teamwork at every level

University exam:
Students split syllabus (model parallel)
Different classes attempt different questions (data parallel)
Exam runs in shifts (pipeline)

Hybrid parallelism

GPT-4 training
Uses:
Data parallelism
Model parallelism
Pipeline parallelism

Combination of all techniques

1 Unlocking the phone
2 Recognizing the images
3 Issue of normal neural network
4 CNN
5 Standard Classification of images
6 Types of Images
7 Colored image
8 Understanding the pixel grid
9 Edge Detection
10 Edge Detection importance
11 Detecting edges
12 Understanding Filter
13 Vertical Edge Detection
14 Horizontal Edge Detection
15 Convolution Operation
16 Kernel
17 Stride and Padding
18 Pooling Layer
19 CNN Architecture
20 Vertical Edge Detection
21 Detection
22 Resultant image
23 Filters
24 What Else Can Convolution Do?
25 How CNNs Use Layers
26 Applications
27 Training CNN
28 Famous CNN Architectures
29 CNN vs ANN vs RNN
30 HPC
Distributed memory
32 Data Parallelism
33 Model Parallelism
34 Pipeline Parallelism
35 Hybrid Parallelism

Data parallelism: “Same model, different data”
Model parallelism: “Different model parts, same data”
Pipeline parallelism: “Assembly line training”
Hybrid: “All together”

AI Needs	HPC Provides
Big data	Parallel processing
Large models	Distributed memory
Fast training	GPUs
Scaling	Clusters

What Runs on HPC Today?
ChatGPT
Weather models
Drug discovery
Autonomous driving
Climate simulations
Satellite image analysis

“AI without HPC is just theory.”
“Parallelism is the real power behind intelligence.”
“Supercomputers don’t think faster — they think together.”

End of Lecture 11

Thank you !